

Docker Image Registry - Complete Guide

A Docker Image Registry is a centralized system for storing, managing, and distributing Docker images. It allows developers to push and pull container images efficiently.

What is Docker Image?

A Docker image is a lightweight, standalone, executable package that includes everything needed to run an application.

What is Docker Registry?

A Docker registry is a storage and distribution system for Docker images. It can be public or private.

Types of Registries

1. Public Registry (Docker Hub) 2. Private Registry 3. Cloud Registries (AWS ECR, GCR, Azure ACR)

Working of Docker Registry

1. Build Image 2. Tag Image 3. Push to Registry 4. Pull from Registry 5. Run Container

Common Commands

```
docker build -t image_name .
docker tag image_name username/image_name
docker push username/image_name
docker pull username/image_name
```

Advantages

Centralized storage, version control, easy sharing, CI/CD integration

Use Cases

Microservices deployment, DevOps pipelines, cloud-native applications